FINDING AN OPEN IMMERSION INTO A U-ADMISSIBLE BLOWUP TARGET AFTER BLOWING UP

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 9197-9222, immediately following the statement at 9170-9195.

Editorial note. One 'small detail omitted' phrase at source line 9221 (X' an open subspace of Z') is retained verbatim and flagged: source line(s) carrying it are listed by the omitted-steps note.

Editorial note. Three cross-chapter pointer families (Morphisms of Spaces, Divisors on Spaces) are recorded in assembly.external_references; the common-blowup lemma is a same-chapter line-indexed stub.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 9221 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Quasi-compact immersions have scheme-theoretically dense image. Morphisms of Spaces, Lemma spaces-morphisms-lemma-quasi-compact-immersion, cited at proof lines 9204-9207. Gives U subset Z scheme-theoretically dense open, where Z is the scheme-theoretic image of U -> P^n_Y.
- Common blowup lemma. Lemma lemma-find-common-blowups of this chapter, cited at proof lines 9208-9211. Produces the diagram X' -> Xbar, Z' -> Z with U-admissible blowups dominating both; the entire conclusion is an unpacking of this diagram.
- U-admissible blowups keep U scheme-theoretically dense. Divisors on Spaces, Lemmas spaces-divisors-lemma-blowing-up-gives-effective-Cartier-divisor and spaces-divisors-lemma-complement-effective-Cartier-divisor, cited at proof lines 9212-9216. U stays scheme-theoretically dense in both X' and Z' after the admissible blowups.
- Proper morphisms are closed. Morphisms of Spaces, Lemma spaces-morphisms-lemma-universally-closed-permanence, cited at proof lines 9217-9219. Z' -> Z -> Y proper makes Z' a closed subspace of Xbar, the pivot for the final inclusion.

Source reference. Original lemma 088Q: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9084–9084. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 085T: focused-seeds/geometry/S9-spaces-blowup-to-find-embedding/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 083S: focused-seeds/geometry/S9-spaces-blowup-to-find-embedding/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

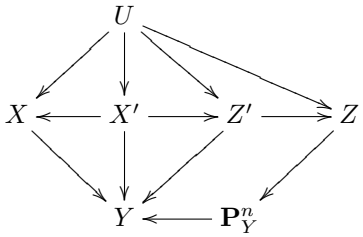
Source reference. Original reference 088G: focused-seeds/geometry/S9-spaces-blowup-to-find-embedding/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NX: focused-seeds/geometry/S9-spaces-blowup-to-find-embedding/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $U \subset X$ be an open subspace. Assume*

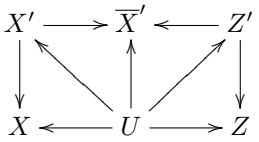
(1) *U is quasi-compact,*
(2) *Y is quasi-compact and quasi-separated,*
(3) *there exists an immersion $U \to \mathbf{P}^n_Y$ over Y,*
(4) *f is of finite type and separated.*

Then there exists a commutative diagram



where the arrows with source U are open immersions, $X' \to X$ is a U-admissible blowup, $X' \to Z'$ is an open immersion, $Z' \to Y$ is a proper and representable morphism of algebraic spaces. More precisely, $Z' \to Z$ is a U-admissible blowup and $Z \to \mathbf{P}^n_Y$ is a closed immersion.

Proof. Let $Z \subset \mathbf{P}^n_Y$ be the scheme theoretic image of the immersion $U \to \mathbf{P}^n_Y$. Since $U \to \mathbf{P}^n_Y$ is quasi-compact we see that $U \subset Z$ is a (scheme theoretically) dense open subspace (Morphisms of Spaces, Lemma 088G). Apply Lemma 088Q to find a diagram



with properties as listed in the statement of that lemma. As $X' \to X$ and $Z' \to Z$ are U-admissible blowups we find that U is a scheme theoretically dense open of both X' and Z' (see Divisors on Spaces, Lemmas 085T and 083S). Since $Z' \to Z \to Y$ is proper we see that $Z' \subset \overline{X}'$ is a closed subspace (see Morphisms of Spaces,

Lemma 04NX). It follows that $X' \subset Z'$ (scheme theoretically), hence X' is an open subspace of Z' (small detail omitted) and the lemma is proved. □